

Based on the training and testing logs you've shared during development, here is a consolidated **Authentix Audio AI Training Results** table.

🎵 Authentix Audio AI Final Training Results

Model Information

Parameter	Value
Model	MelodyMachine/Deepfake-audio-detection-V2
Backbone	Wav2Vec2
Framework	PyTorch + Hugging Face Transformers
Training Strategy	Transfer Learning
Fine-Tuning	Three-Phase Progressive Unfreezing
Mixed Precision	✅ Enabled
Device	NVIDIA RTX 3050 Laptop GPU (CUDA)

Training Configuration

Parameter	Value
Total Parameters	94,569,090
Phase 1 Trainable Parameters	197,378
Backbone Frozen	✅ Yes
Optimizer	AdamW
Scheduler	CosineAnnealingLR
Loss Function	CrossEntropyLoss
Label Smoothing	✅ Enabled
Gradient Clipping	✅ Enabled
TensorBoard	✅ Enabled

Parameter	Value
Checkpointing	Enabled
Resume Training	Supported

Dataset Summary

Metric	Value
Raw Audio Files	1,990
Successfully Processed	1,838
Rejected Files	152
Training Samples	1,466
Validation Samples	183
Test Samples	189
Real Samples	840
Fake Samples	998

Training Results

Phase	Training Strategy	Validation Accuracy	F1 Score	ROC-AUC
Phase 1	Train Classifier Only	63.9%	57.7%	0.664
Phase 2	Fine-Tune Last Encoder Layers	63.9%	57.7%	0.669
Phase 3	Fine-Tune Entire Wav2Vec2	100.0%	100.0%	1.000

Final Test Results

Metric	Score
Accuracy	100.00%
Balanced Accuracy	100.00%
Precision	100.00%
Recall	100.00%
F1 Score	100.00%
ROC-AUC	1.0000
False Positive Rate	0.0000
False Negative Rate	0.0000

Confusion Matrix

Actual \ Predicted	Real	Fake
Real	88	0
Fake	0	101

Total Test Samples:

189

Correct Predictions:

189 / 189

Training Features Implemented

Feature	Status
Automatic Hugging Face Model Download	

Feature	Status
Automatic Processor Download	✓
Three-Stage Transfer Learning	✓
Mixed Precision (AMP)	✓
TensorBoard Logging	✓
Early Stopping	✓
Gradient Clipping	✓
Automatic Checkpoint Saving	✓
Resume Training	✓
Automatic Best Model Restoration	✓
CUDA Support	✓

Generated Artifacts

Checkpoints

- ✓ `best_model.pth`
- ✓ `last_model.pth`
- ✓ `optimizer.pt`
- ✓ `scheduler.pt`
- ✓ `training_state.json`

Reports

- ✓ `classification_report.txt`
- ✓ `evaluation_report.md`
- ✓ `test_results.json`
- ✓ `predictions.csv`

Visualizations

- ✓ `confusion_matrix.png`
- ✓ `roc_curve.png`

-  `precision_recall_curve.png`
-  Training Loss Curve
-  Training Accuracy Curve
-  TensorBoard Logs

Overall Assessment

Category	Rating
Audio Preprocessing Pipeline	★★★★★
Dataset Management	★★★★★
Hugging Face Integration	★★★★★
Training Pipeline	★★★★★
Checkpoint Management	★★★★★
Evaluation Pipeline	★★★★★
ONNX Readiness	★★★★★

Final Notes

The Audio AI module is **feature-complete** from a software engineering perspective:

- **Preprocessing:** Complete
- **Training:** Complete
- **Testing & Evaluation:** Complete

The one remaining technical recommendation is to **evaluate the model on an external dataset** (such as **ASVspoof** or **WaveFake**) before treating the reported **100% test accuracy** as a measure of real-world performance. Your current test set comes from the same dataset family used for fine-tuning, so an external benchmark is the best way to demonstrate generalization. After that, the next logical steps are **ONNX export** and **integration into Authentix's multimodal fusion pipeline**.

